Phan Đức Anh

Java Backend Developer

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OBJECTIVE

Short-term goal: Join as a Fresher Java Backend Developer to contribute to real-world projects, deepen my expertise in Spring Boot ecosystem, and learn best practices from experienced engineers.
Long-term goal: Advance to a Mid-level Backend Developer within 2-3 years, mastering microservices architecture, cloud technologies, and taking ownership of complex system components.

EDUCATION

FPT University Ha Noi 2022 - 2026

SOFTWARE ENGINEERING (SEMESTER 9/9)

Specialization: Korea SE
GPA: 3.27

PROJECTS

SatoriNihongo - *Japanese Learning Support Application* 01/2026 - 05/2026

Role: Back-end Developer

- **Team size:** 5
- **Brief**: An AI-powered Japanese-language learning platform (JLPT N5–N1) built on a Spring Boot multi-module architecture (13 feature modules). Combines agentic RAG curriculum ingestion, a document-grounded AI tutor, automated pronunciation and roleplay assessment, and adaptive practice — backed by PostgreSQL + pgvector and Redis.
- **Programing Languages:** Java
- **Technologies**: Spring Boot 3.3.5, Spring AI, Spring Security, JWT, Spring Data JPA, Spring Application Events (event-driven cross-module)
- **Tools**: PostgreSQL + pgvector, Redis, OpenAI (GPT-4o), Google Document AI, Azure Speech / SpeechSuper, Cloudinary, Resend, Firebase FCM, Docker, Git
- **Key features:**
  - Agentic RAG curriculum import — scanned PDF ingestion via Google Document AI OCR with GPT-4o Vision fallback; heading-aware chunking, embedding into pgvector, and Corrective/Self-RAG grounding verification with confidence-scored re-extraction to block hallucinations.
  - Document-grounded AI tutor — RAG chat scoped per curriculum via Spring AI advisor chain (semantic intent router + relevance/grounding checks); hybrid retrieval (dense HNSW + sparse pg_trgm fused with RRF) plus HyDE for higher recall.
  - Real-time streaming — chat responses streamed to client over Server-Sent Events (SSE).
  - Speaking assessment — pronunciation scoring (Azure Speech / SpeechSuper) and AI roleplay sessions with mission-based, weighted evaluation.
  - Adaptive practice — AI-generated exercises with an EMA-based mastery model and anti-repeat windowing.
  - Secure stateless auth — JWT with dual access/refresh secrets and rotating refresh tokens, Redis-backed fixed-window per-IP rate limiting, and account lockout on failed attempts.
  - Modular architecture — schema-per-module PostgreSQL design, decoupled cross-module communication via query ports + application events; email (Resend), media (Cloudinary), and push notifications (FCM).
- **Github:** https://github.com/ThuyetTVHE186110/satori-learning-support-application-backend

Cognitive Lab — AI-Powered Document Learning Platform 05/2026 - Now

Role: Solo Developer

- **Team size**: 1
- **Brief**: A full-stack RAG (Retrieval-Augmented Generation) system that allows users to upload documents and interact with them via AI chat. Features include chunked file upload with async processing, vector-based semantic search, AI-generated learning roadmaps, progressive quiz generation, and real-time progress streaming via SSE.
- **Programming Languages:** Java
- **Techs**: Spring Boot, Spring AI, Spring Security, JWT, Spring Data JPA, RabbitMQ, Redis
- **Tools**: PostgreSQL + pgvector, MinIO, Ollama, Docker, Git
- **Key features**:
  - Chunked upload pipeline: *MinIO → RabbitMQ → Apache Tika → TokenTextSplitter → pgvector embedding*
  - RAG chat scoped per document using QuestionAnswerAdvisor + pgvector cosine similarity
  - Async quiz & roadmap generation via RabbitMQ with live SSE streaming to client
  - Stateless JWT auth with rotating refresh tokens and Redis-backed rate limiting (fixed-window per IP)
- **Github:** https://github.com/anhpd912/cognitive-lab-server

Personal Portfolio — Developer Portfolio Website 06/2026 - Now

Role: Solo Developer

- **Team size**: 1
- **Brief**: A single-page personal portfolio built on Next.js 16 (App Router) with a hand-built neobrutalist design system. Showcases skills, projects, and contact info; statically exported and hosted on GitHub Pages.
- **Programming Languages**: TypeScript
- **Techs**: Next.js 16, React 19, Tailwind CSS v4
- **Tools**: ESLint, Turbopack, next/font, GitHub Pages, Git
- **Key features**:
    - Custom CSS design-token system (color, spacing, typography, motion, hard-shadow) driving a consistent neobrutalist look
    - Component-based section architecture (hero, skills, projects, about, contact) with reusable UI primitives (button, card, tag)
    - Viewport-aware lazy rendering via content-visibility for fast first paint
    - Fully responsive with accessible high-contrast focus states and self-hosted next/font (Space Grotesk)
    - Static export deployed to GitHub Pages
- **Live**: https://anhpd912.github.io/portfolio/
- **Github**: https://github.com/anhpd912/portfolio

TECHNICAL SKILLS

Programing languages	Java, Python, Javascript, HTML/CSS, C
Framework & Library	Spring Boot React.js, Next.js
Tools & Platform	Postman/JUnit Git/Github, Docker Claude Code
Databases	SQL Server, MySQL, PostgreSQL, MongoDB, Redis
Soft Skills	**English** (Intermediate) & **Korean** (Intermediate) Teamwork, problem-solving & self-learning

CERTIFICATIONS

(English) TOEIC L&R - 750	2025
(Korean) TOPIK 4	2025
Entrepreneurship: Launching an Innovative Business (Coursera)	2025
CertNexus Certified Ethical Emerging Technologist (Coursera)	2024
Web Design for Everybody: Basics of Web Development & Coding (Coursera)	2023
Software Development Lifecycle (Coursera)	2022
Academic Skills for University Success (Coursera)	2022

ACTIVITIES

Member Of Expertise Department 04/2024 - Now

JS Club - Japanese Software Engineers

Actively participate in club activities, join algorithm competitions, and coding competitions within the club.